BM4 Implicit 2 with Hairer's Symmetric Projection

Simultaneous Newton formulation for output and multiplier

Schur-complement equivalent to the reduced projected BM4 method

1 Purpose and relation to BM4 implicit 1

`BM4Implicit2` computes the same exact projected physical map as `BM4Implicit1`, but retains the duplicated output state as a Newton unknown. For one guiding-centre particle, implicit 1 solves only for a two-component multiplier, whereas implicit 2 solves simultaneously for four duplicated output components and two multiplier components. Its Newton system is therefore 6×6.

Both methods surround one complete twelve-stage BM4 cycle with Hairer's symmetric projection. Neither method projects after the individual BM4 stages; that different operation belongs to `ProjectedBM4Composition`.

2 Doubled state and projection operators

For the physical guiding-centre state $z = (X, Y) \in \mathbb{R}^2$, define

$$\mathcal{Y} = (u, v) \in \mathbb{R}^4, \qquad \mathcal{M} = \{(u, v) : u = v\}.$$

The relevant constant matrices are

$$E = \begin{pmatrix} I_2 \\ I_2 \end{pmatrix}, \qquad G = (I_2 \ \ -I_2), \qquad N = G^T = \begin{pmatrix} I_2 \\ -I_2 \end{pmatrix}.$$

Consequently,

$$GE = 0, \qquad GN = 2I_2.$$

The input Ez_n lies on the physical diagonal and $N\mu = (\mu, -\mu)$ is normal to it.

3 The BM4 base map

Let $D_{s,t}$ denote the direct GC extended map and $D^*_{s,t}$ its adjoint. The direct map updates the second copy, then the first copy, and then applies the exact harmonic coupling. The adjoint applies the coupling first and then performs the triangular updates in reverse order.

The BM4 coefficients are

$$\begin{aligned} a_1 &= 0.0792036964311957, & a_2 &= 0.1303114101821663, \\ a_3 &= 0.2228614958676077, & a_4 &= -0.3667132690474257, \\ a_5 &= 0.3246481886897062, & a_6 &= 0.1096884778767498. \end{aligned}$$

With

$$(b_1, \ldots, b_{12}) = (a_1, a_2, a_3, a_4, a_5, a_6, a_6, a_5, a_4, a_3, a_2, a_1),$$

the implementation alternates adjoint and direct stages. Define

$$\tau_0 = t_n, \qquad \tau_j = t_n + h \sum_{k=1}^{j} b_k.$$

Then

$$\mathcal{Y}^{(j)} = \begin{cases} D^{*}_{b_j h,\,\tau_{j-1}}(\mathcal{Y}^{(j-1)}), & j \text{ odd}, \\ D_{b_j h,\,\tau_j}(\mathcal{Y}^{(j-1)}), & j \text{ even}, \end{cases}$$

and

$$\boxed{\Phi^{\mathrm{BM4}}_{h,t_n}(\mathcal{Y}^{(0)}) = \mathcal{Y}^{(12)}.}$$

The sum of the twelve coefficients is one. Their palindromic order and the direct/adjoint pairing define a symmetric fourth-order map, including the negative fourth coefficient.

4 Simultaneous projection equations

Start from the physical input $\mathcal{Y}_n = Ez_n$ and pre-correct it with the unknown multiplier:

$$\widehat{\mathcal{Y}}_n(\mu) = Ez_n + N\mu.$$

BM4 implicit 2 retains the diagonal output $\mathcal{Y}_{n+1}$ as a separate unknown and imposes

$$d(\mathcal{Y}_{n+1}, \mu) := \mathcal{Y}_{n+1} - N\mu - \Phi^{\mathrm{BM4}}_{h,t_n}(Ez_n + N\mu) = 0, \tag{1}$$

$$g(\mathcal{Y}_{n+1}) := G\mathcal{Y}_{n+1} = 0. \tag{2}$$

If $\mathcal{Y}_{n+1} = (y_u, y_v)$ and $\Phi^{\mathrm{BM4}}_{h,t_n}(Ez_n + N\mu) = (u_f(\mu), v_f(\mu))$, the residual is

$$d_u = y_u - \mu - u_f(\mu), \qquad d_v = y_v + \mu - v_f(\mu), \qquad g = y_u - y_v.$$

Equations (1)–(2) express both the projected BM4 step and the final diagonal constraint without eliminating any output variable.

5 Simultaneous Newton matrix

Set

$$J_{\mathrm{BM4}} = D\Phi^{\mathrm{BM4}}_{h,t_n}(Ez_n + N\mu).$$

Since E, G, and N are constant,

$$D_{\mathcal{Y}_{n+1}} d = I_4, \qquad D_\mu d = -(I_4 + J_{\mathrm{BM4}})N,$$

$$D_{\mathcal{Y}_{n+1}} g = G, \qquad D_\mu g = 0.$$

At iteration k, the simultaneous Newton correction solves

$$\boxed{\begin{pmatrix} I_4 & -(I_4 + J_{\mathrm{BM4}})N \\ G & 0 \end{pmatrix} \begin{pmatrix} \Delta\mathcal{Y}_{n+1} \\ \Delta\mu \end{pmatrix} = -\begin{pmatrix} d \\ g \end{pmatrix}.} \tag{3}$$

The updates are

$$\mathcal{Y}^{(k+1)}_{n+1} = \mathcal{Y}^{(k)}_{n+1} + \Delta\mathcal{Y}_{n+1}, \qquad \mu^{(k+1)} = \mu^{(k)} + \Delta\mu.$$

No inverse is formed explicitly. For N physical particles, the current code assembles a dense $6N \times 6N$ system.

6 Exact SymPy expression for the BM4 Jacobian

Write $W(z,t) = D_z f(z,t)$. For $s_j = b_j h$, coupling frequency ω, and

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix},$$

define the two coupling blocks

$$H_j = \frac{I_2 + R(2\omega s_j)}{2}, \qquad K_j = \frac{I_2 - R(2\omega s_j)}{2}.$$

The Jacobian of the exact linear coupling is therefore

$$C_j = \begin{pmatrix} H_j & K_j \\ K_j & H_j \end{pmatrix}.$$

For an even, direct stage j, let

$$\widetilde{v}_j = v^{(j-1)} + s_j f(u^{(j-1)}, \tau_j),$$

and set

$$U_j = W(u^{(j-1)}, \tau_j), \qquad V_j = W(\widetilde{v}_j, \tau_j), \qquad L_j = I_2 + s_j^2 V_j U_j.$$

SymPy multiplication of the two shears and the final coupling gives the exact 4×4 direct-stage Jacobian

$$\boxed{\begin{aligned} \mathcal{D}_j &= C_j \begin{pmatrix} L_j & s_j V_j \\ s_j U_j & I_2 \end{pmatrix} \\ &= \begin{pmatrix} H_j L_j + s_j K_j U_j & s_j H_j V_j + K_j \\ K_j L_j + s_j H_j U_j & s_j K_j V_j + H_j \end{pmatrix}. \end{aligned}}$$

For an odd, adjoint stage, first write the coupled input as

$$\begin{pmatrix} \overline{u}_j \\ \overline{v}_j \end{pmatrix} = C_j \begin{pmatrix} u^{(j-1)} \\ v^{(j-1)} \end{pmatrix}, \qquad u^{(j)} = \overline{u}_j + s_j f(\overline{v}_j, \tau_{j-1}).$$

Now define

$$U_j = W(\overline{v}_j, \tau_{j-1}), \qquad V_j = W(u^{(j)}, \tau_{j-1}), \qquad L_j = I_2 + s_j^2 V_j U_j.$$

The exact adjoint-stage Jacobian is

$$\boxed{\begin{aligned} \mathcal{A}_j &= \begin{pmatrix} I_2 & s_j U_j \\ s_j V_j & L_j \end{pmatrix} C_j \\ &= \begin{pmatrix} H_j + s_j U_j K_j & K_j + s_j U_j H_j \\ s_j V_j H_j + L_j K_j & s_j V_j K_j + L_j H_j \end{pmatrix}. \end{aligned}}$$

The order of every U_j, V_j, H_j, and K_j factor is essential: these 2×2 matrices do not commute in general.

Let $\mathcal{A}_j[c]$ or $\mathcal{D}_j[c]$ mean that the corresponding stage duration is $s_j = ch$. Substitution of the palindromic BM4 coefficients and application of the chain rule gives the requested explicit expression

$$\boxed{\begin{aligned} J_{\mathrm{BM4}} = {}& \mathcal{D}_{12}[a_1]\mathcal{A}_{11}[a_2]\mathcal{D}_{10}[a_3]\mathcal{A}_9[a_4]\mathcal{D}_8[a_5]\mathcal{A}_7[a_6] \\ & \times \mathcal{D}_6[a_6]\mathcal{A}_5[a_5]\mathcal{D}_4[a_4]\mathcal{A}_3[a_3]\mathcal{D}_2[a_2]\mathcal{A}_1[a_1]. \end{aligned}}$$

This factorization is exact, not a finite-difference approximation. The reproducible calculation in `bm4_jacobian_sympy.py` uses SymPy 1.14, verifies both displayed stage identities, and preserves noncommutative block order. Even before the internal stage blocks are expanded, the fully expanded twelve-factor product contains 2048 ordered monomials in each final 2×2 block, or 8192 in total. The factorized expression above is therefore the explicit form suitable for analysis and implementation.

Centered-difference production evaluation

The current implementation evaluates this exact derivative by centered differences of the complete doubled map:

$$[J_{\mathrm{BM4}}]_{:,j} \approx \frac{\Phi^{\mathrm{BM4}}_{h,t_n}(\widehat{\mathcal{Y}} + \varepsilon_j e_j) - \Phi^{\mathrm{BM4}}_{h,t_n}(\widehat{\mathcal{Y}} - \varepsilon_j e_j)}{2\varepsilon_j},$$

$$\varepsilon_j = \eta \max(1, |\widehat{Y}_j|), \qquad \eta = \sqrt[3]{\epsilon_{\mathrm{mach}}}.$$

This choice reuses the exact production BM4 cycle and avoids maintaining a second twelve-stage tangent implementation. Its trade-off is the additional map evaluations and a finite-difference floor in the Newton matrix.

7 Schur complement and equivalence to implicit 1

Eliminating $\Delta\mathcal{Y}_{n+1}$ from (3) gives

$$\mathcal{K} = G(I_4 + J_{\mathrm{BM4}})N = 2I_2 + GJ_{\mathrm{BM4}}N. \tag{4}$$

The reduced residual is

$$R(\mu) = G\Phi^{\mathrm{BM4}}_{h,t_n}(Ez_n + N\mu) + 2\mu.$$

Therefore

$$R'(\mu) = \mathcal{K}.$$

Moreover, applying G to (1) gives

$$Gd = g - R(\mu).$$

The Schur complement of the simultaneous system is consequently

$$\boxed{\mathcal{K}\Delta\mu = -R(\mu),}$$

which is exactly the reduced Newton correction used by `BM4Implicit1`. The formulations describe the same nonlinear root; they differ only in which variables are retained in the linear system.

8 Initial iterate and stopping rule

The implementation starts from

$$\mu^{(0)} = 0, \qquad \mathcal{Y}^{(0)}_{n+1} = \Phi^{\mathrm{BM4}}_{h,t_n}(Ez_n).$$

This makes $d = 0$ initially, while the diagonal defect g drives the first correction. For

$$s_z = \max(1, \|z_n\|_\infty),$$

the iteration stops when

$$\max(\|d\|_\infty, \|g\|_\infty) \le \mathrm{atol} + \mathrm{rtol}\, s_z.$$

At convergence, the neutral physical representative is

$$z_{n+1} = \frac{y_u + y_v}{2}.$$

9 Symmetry, symplecticity, and order

At the exact root, equations (1)–(2) are the same symmetric projection condition as in implicit 1. If the root is locally unique, reversing the BM4 step and changing the sign of the multiplier yields

$$\Psi_{-h,t_{n+1}} \circ \Psi_{h,t_n} = I.$$

The projected method retains fourth-order accuracy under the standard regularity conditions. Its projection multiplier is expected to scale as

$$\|\mu_h\|_\infty = O(h^5).$$

If the doubled BM4 cycle preserves the extended symplectic form, Hairer's exact symmetric projection induces a symplectic physical map. Finite residual tolerances, centered-difference derivatives, and linear-solve round-off set the observed structural-error floor.

10 Reproducible comparison with BM4 implicit 1

The numerical audit uses:

- random potential amplitude 0.3, maximum wave number 8, grid 32×32, seed 27, interpolation order 5;
- $z_0 = (1, 1.2)$, $\rho = 0.3$, coupling frequency $\omega = \pi/8$;
- time interval $[0, 0.4]$, absolute and relative Newton tolerances 10^{-15}, and at most 12 iterations;
- centered-difference scale $\eta = \sqrt[3]{\epsilon_{\text{mach}}}$.

h	$\max \lvert z^{(1)} - z^{(2)} \rvert$	$\max \|\mu\|_\infty$	max residual	max it.
0.200	6.66×10^{-16}	6.0004×10^{-10}	1.83×10^{-15}	1
0.100	3.33×10^{-16}	2.1510×10^{-11}	1.31×10^{-15}	1
0.050	2.22×10^{-16}	7.1747×10^{-13}	1.00×10^{-15}	1
0.025	2.22×10^{-16}	2.3095×10^{-14}	1.44×10^{-15}	1

The superscripts (1) and (2) denote the reduced and simultaneous formulations. Their state differences remain at round-off. Against a $h = 0.003125$ reference, the reduced formulation gives observed global orders 4.013, 4.002, and 3.990; the simultaneous formulation produces the same states to the precision shown. The multiplier orders 4.802, 4.906, and 4.957 approach the expected value five.

11 Implementation correspondence

The production implementation follows the mathematical blocks directly:

- `BM4Implicit2` selects the simultaneous solver;
- `_advance_composition` evaluates the complete BM4 base map;
- `_bm4_map_jacobian` evaluates J_{BM4};
- `_solve_simultaneous_projected_bm4_step` assembles (1)–(3);
- nonlinear diagnostics record iterations, final residuals, and multiplier norms for every main integration step;
- the reusable comparison study runs both formulations on identical grids and reports aligned physical-flow diagnostics.